

MA2040 Jan-Apr 2024

Problem Sheet 2

1. Let P be a probability measure and A and B be events. Show that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
2. Prove that, for a probability measure P and events $E_1, \dots, E_n$,

$$P\left(\bigcup_{i=1}^{i=n} E_i\right) \leq \sum_{i=1}^{i=n} P(E_i).$$

3. Let $S = \{s_1, \dots, s_m\}$. Suppose that $p_1, \dots, p_m$ are non-negative numbers such that $p_1 + \dots + p_m = 1$. For $A \subset S$, define $P(A) = \sum_{j:s_j \in A} p_j$. Show that P satisfies all axioms of a probability law.
4. Suppose that a bag contains 100 slips such that in 95 slips the letter 'A' is written and in 5 slips letter 'B' is written. Three slips are drawn at random with replacement. What is the probability of getting two or more A's?
5. Suppose that a fair coin is tossed until a head is obtained. Describe the sample space. What is the probability that exactly k tosses are required?
6. Let B be an event such that $P(B) > 0$. For an event A , define $P_B(A) = P(A|B)$. Show that P_B satisfies all axioms of a probability law.
7. Prove that if the events A and B are independent, then A^c and B^c are also independent.
8. If an aircraft is present in a certain area, then a radar correctly registers its presence with probability 0.99. If it is not present, then the radar falsely indicates its presence with probability 0.10. Suppose the aircraft is present in the area with probability 0.05. Find the probability of
 - (a) false detection, that is, the radar registers the presence of aircraft even though it is not present
 - (b) missed detection, that is, the radar does not register the presence of aircraft even though

it is present

(c) presence of aircraft given that the radar has registered its presence.

9. We roll a fair die. If the result is 1 or 2, we roll once more. Otherwise, we stop. What is the probability that the sum of the two rolls is at least 4?
10. An internet service provider has provided 50 modems to serve the needs of 300 customers. It is estimated that at a given time each customer will need a connection with probability 0.1 independent of other customers. What is the probability that there are more customers needing a connection than there are modems?
11. Consider the following guessing problem. Suppose Bob thinks of a random integer X between 1 to M with a probability distribution P_X , i.e., $P_X(x) := P(X = x)$ for $x = 1, \dots, M$. Alice tries to guess it with a probability distribution $\hat{P}$ by asking questions only of the form, is $X = x$, with every guess being independent of her previous guesses. What is the probability that she will be able to guess it in exactly k trials?